

Documentation: Methods Used to Access Intel NUC Screen from Raspberry Pi

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Project Goal: Explore multiple methods to access and control an Intel NUC (headless) from a Raspberry Pi using only the RPi screen, ideally without prior knowledge of the NUC's IP or installing anything on it.

Use Case

Access and control a headless NUC system through a Raspberry Pi screen — including full screen sharing, mouse, and keyboard control — under constraints such as:

- No additional hardware setup
- No installation on the NUC
- No prior knowledge of NUC's IP in some cases

Methods Attempted

1. Remote Desktop via Remmina (RDP)

- **Setup:** Installed remmina on the Raspberry Pi.
- **Requirement:** IP address of the NUC, RDP service enabled on NUC.
- **Result:** ✅ Successfully accessed full NUC screen with keyboard/mouse control.
- **Limitation:** Required NUC's IP and preconfigured RDP (which may not always be available).

2. X2Go (NX Protocol)

- **Setup:** Installed X2Go Client on RPi, attempted to connect to NUC.
- **Requirement:** IP, SSH access, and X2Go Server on NUC.
- **Result:** ❌ Screen opened briefly but showed black screen and exited.
- **Limitation:** Needed X2Go server installed on the NUC.

3. SSH with X11 Forwarding

- **Command:** `ssh -X ubuntu@<NUC_IP>`
- **Goal:** Run GUI applications from NUC, rendered on the Pi.
- **Result:** ✅ Worked for individual apps like gedit, nautilus.
- **Limitation:** ❌ Did not provide full desktop control, only app-level.

4. USB HID Gadget Mode

- **Goal:** Make Raspberry Pi act like a keyboard/mouse using uinput or rutomatrix.
- **Result:** ✅ Successful control of NUC (e.g., send keystrokes).
- **Limitation:** ❌ No screen feedback; required HDMI capture to see NUC screen.

5. HDMI Capture + Virtual HID

- Setup: NUC HDMI output connected to Raspberry Pi (via capture device).
- Goal: View NUC's screen as a video feed on Pi while sending keyboard/mouse.
- Result: ☒ Fully working — visual + input control.
- Limitation: Required hardware HDMI connection, not purely network-based.

6. NoMachine (NX Protocol Alternative to RDP/VNC)

- Setup: Installed NoMachine client on Raspberry Pi.
- Usage:
 - Scanned LAN for available NUC if IP not known.
 - Manually entered known NUC IP (172.16.38.56).
- Result: ☒ Successfully accessed NUC screen with full mouse/keyboard control.
- Limitation: Required NoMachine server already running on NUC.

7. Intel AMT / vPro (Port 16992 Web Access)

- Goal: Use browser-based remote access to BIOS-level control.
- Test: `nmap -p 16992 <NUC_IP>` and browser access.
- Result: ☒ Port closed, AMT not enabled on NUC.
- Limitation: Required hardware support and BIOS setup in advance.

8. Network Scanning with nmap

- Use: Discover the NUC's IP address when unknown.
- Command:
 - `bash`
 - `CopyEdit`
 - `nmap -sn 172.16.38.0/24`
 - `nmap -p 4000 172.16.38.0/24`
- Result: ☒ Helped identify Intel NUC by MAC/vendor or open ports.
- Limitation: Still requires NUC to be powered on and connected to same network.

Summary Table

Method	Worked?	Requires IP?	Requires Software on NUC?	Screen Access	Input Control
Remmina (RDP)	✓	✓ Yes	✓ Yes	✓ Full desktop	✓ Yes
X2Go	✗	✓ Yes	✓ Yes	✗ Failed to show screen	✗
SSH + X11	✓	✓ Yes	✗ No	✗ App-only	✓ App-level
USB HID / Gadget	✓	✗ No	✗ No	✗ Need HDMI	✓ Yes
HDMI + USB HID	✓	✗ No	✗ No	✓ Yes	✓ Yes
NoMachine	✓	✓ Yes (or auto-detect)	✓ Yes	✓ Full desktop	✓ Yes
Intel AMT	✗	✓ Yes	✗ No	✗ Not enabled	✗
nmap IP Scan	✓	✗ No	✗ No	⊘ (Discovery only)	⊘

Key Learnings

- IP discovery tools (like nmap) are crucial when managing unknown headless devices.
- NoMachine provides an excellent, lightweight alternative to RDP/VNC if preinstalled.
- In total isolation (no IP, no RDP, no installs), only hardware-based solutions (like HDMI + USB HID) are reliable.